TEXORA · LECTURE SHEET

Sewing Technology

Stitch and seam as standardised engineering objects: ISO 4915 stitch classes, ISO 4916 seam classes, the lockstitch formation cycle, seam strength and the feeding and penetration that decide whether a seam succeeds.

TE-3107 · APPAREL MANUFACTURING TECHNOLOGY-II · TOPIC 1 · LEVEL 3 TERM 1

1. What sewing is, in engineering terms

Sewing joins fabric plies with a flexible thread structure driven by a needle. Unlike welding or bonding, a sewn seam can flex, stretch within limits, be unpicked and repaired, and join dissimilar materials — which is why, after two centuries of alternatives, it remains the dominant joining method in apparel. This course treats sewing as a system: **stitch formation** (how threads are interlaced), **seam construction** (how plies are arranged), **feed and penetration** (how material moves and is pierced), and the **quality window** — stitch density, tension balance and needle condition — inside which the system produces a saleable seam.

2. The stitch: ISO 4915 classification

Stitch types are standardised in **ISO 4915**. The first digit gives the class; the full three-digit number identifies the exact thread arrangement.

CLASS	NAME	CHARACTER AND TYPICAL USE
100	Single-thread chainstitch	One needle thread looped through itself. Easy to unravel; basting, button sewing, blind hemming (103).
200	Hand stitch	Thread passes in and out by hand or hand-simulating machines; decorative saddle stitching.
300	Lockstitch	Needle thread interlaces with a bobbin thread inside the material. 301 is the general workhorse; 304 zigzag.
400	Multi-thread chainstitch	Needle thread(s) looped with a looper thread below. 401 two-thread chain: strong, elastic, inseams.

500	Overedge chainstitch	Threads wrap the fabric edge, trimming it as they sew. 504 three-thread overlock; 516 five-thread safety stitch (401 + 504 combined).
600	Covering chainstitch (flatlock/coverstitch)	Needle threads plus a cover thread on top and looper below; flat, elastic seams for knit hems (602, 605, 607).

3. Formation of the lockstitch (301)

The lockstitch is the reference stitch of the industry, so its cycle must be understood exactly (Figure 1). One revolution of the machine main shaft produces one stitch.

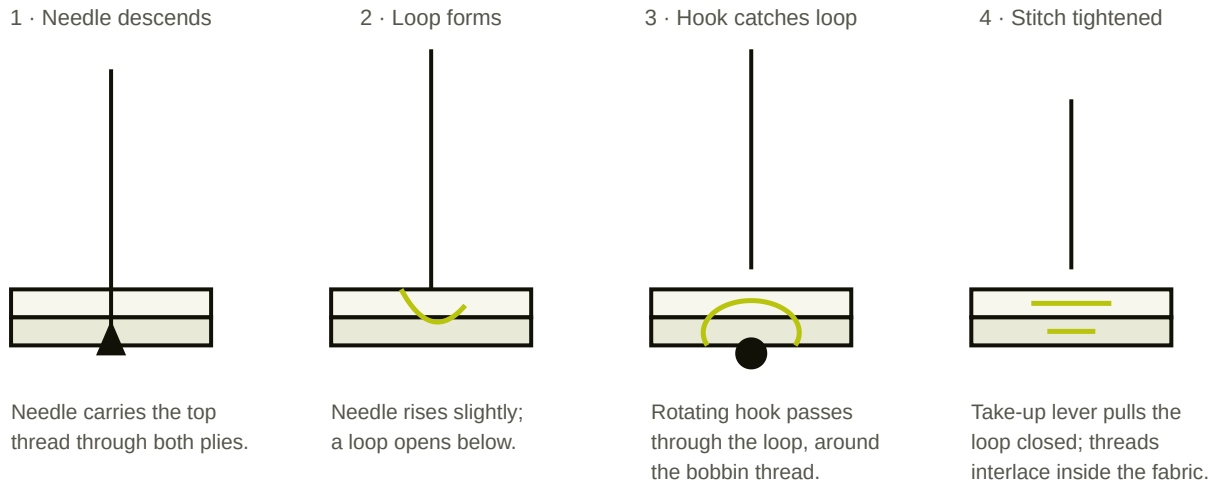


Figure 1 — Formation of the lockstitch (ISO stitch type 301) in four stages. The needle thread and bobbin thread interlace in the middle of the material.

1. **Penetration.** The needle, threaded from its long-groove side, descends and carries the needle thread through both plies to below the needle plate.
2. **Loop formation.** Just after bottom dead centre the needle rises about 1.8–2.2 mm. The thread, held by friction against the fabric, cannot rise with it, so a loop opens on the scarf side of the needle.
3. **Interlacing.** The point of the rotating hook enters the loop and carries it around the bobbin case, so the needle thread encircles the bobbin thread.
4. **Tightening.** The thread take-up lever rises sharply, pulls the loop closed and draws the interlacing point into the middle of the material. The feed dog then advances the fabric by one stitch length while the needle is clear.

A correct 301 shows the same appearance on both faces, with the knot buried mid-ply. If the knot sits on top, bobbin tension is too high (or needle tension too low); if it sits below, the reverse. **Stitch density** is expressed as stitches per cm (or per inch); typical apparel work uses 4–5 stitches/cm (10–13

SPI) on wovens — denser on light fabrics for strength per unit length, coarser on heavy fabrics to limit needle damage and puckering.

4. The seam: ISO 4916 classification

A seam is the junction itself; a stitch is only the thread pattern that fastens it. **ISO 4916** groups seams into eight classes by ply arrangement:

CLASS	CONSTRUCTION AND EXAMPLES
1 — Superimposed	Plies stacked face to face, stitched near the edge: SSa plain seam (side seams), SSaa with a second row.
2 — Lapped	Plies overlap; includes the felled seam LSc (shirt side seam, jeans) where the raw edge is enclosed.
3 — Bound	A binding strip encloses the edge (necklines, armholes of knitwear).
4 — Flat	Edges butt or interlock without overlap and are covered by stitching (flatlock in underwear, sportswear).
5 — Decorative	Stitching that ornaments a single ply (pin tucks, topstitching effects).
6 — Edge finishing	Stitching that neatens a single edge (hemming EFb blind hem, overedged edge).
7 — Joining separate items	Attaching a distinct component: waistband, cuff, elastic, label.
8 — Single-ply construction	Seams formed on one ply alone, e.g. a folded belt loop.

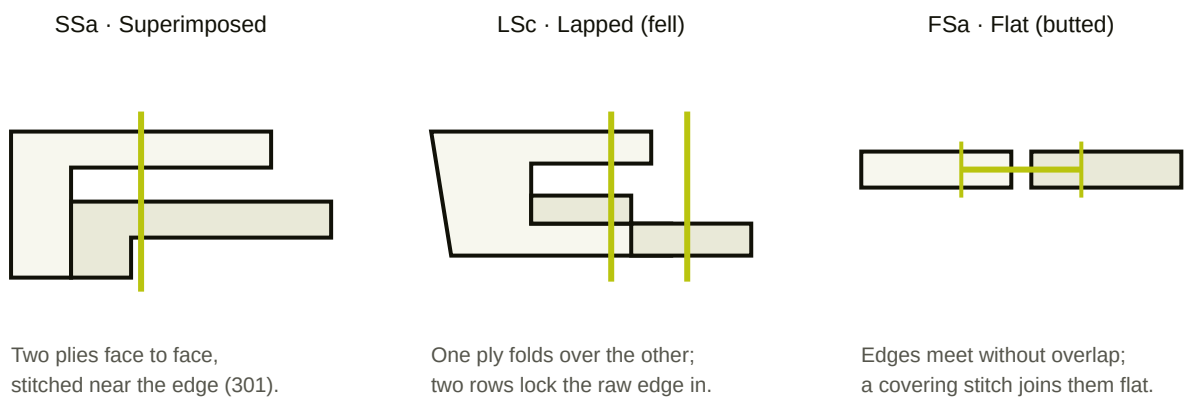


Figure 2 — Three seam constructions in cross-section (ISO 4916): superimposed (Class 1), lapped (Class 2) and flat (Class 4). Green lines are the stitching rows.

5. Seam strength, elasticity and failure

A seam fails in three ways: the **thread breaks**, the **fabric tears** beside the stitch line (needle damage, weak yarns), or the seam **grins/slips** — yarns slide apart under transverse load, opening holes beside the stitching, the classic failure of loose woven trousers loaded across the seat seam. Seam efficiency compares seam strength with fabric strength:

$$\text{Seam efficiency (\%)} = (\text{seam breaking force} \div \text{unsewn fabric breaking force}) \times 100$$

Worked example. A shirting fabric breaks at 320 N; its side seam breaks at 256 N. Efficiency = 256 ÷ 320 × 100 = **80%** — a good result (70–85% is typical for well-made seams; above ~90% the fabric beside the seam usually becomes the weak point). Seam elasticity must match the garment: a 301 seam extends only about 3–5% before thread rupture, while a 401 chain extends roughly 15–20% and a 504 overedge more still — the reason knit garments are never lockstitched on seams that must stretch.

6. Feeding, penetration and needle action

The common **drop feed** uses a feed dog rising through the needle plate, clamping the fabric against the presser foot and moving it one stitch length per cycle. Its weakness is ply shift: the lower ply is driven positively while the upper ply is dragged by friction, so long seams (trouser inseams) can end with the plies displaced — controlled by compound/unison feed and walking feet (Topic 2). Needle penetration force rises with fabric thickness, density and finish; a blunt or wrongly pointed needle raises it further, heating the needle (synthetic fabrics can melt onto it) and cutting yarns instead of parting them. Ball-point needles part knit loops; sharp/set points pierce wovens cleanly; cutting points are reserved for leather.

Glossary

TERM	MEANING
Stitch (ISO 4915)	The thread configuration formed by one needle penetration cycle and its interlacing with other threads.
Seam (ISO 4916)	The junction of plies produced by sewing, classified by how the plies are arranged.
Seam efficiency	Seam breaking force as a percentage of the unsewn fabric breaking force.
Scarf	The recess above the needle eye on the hook side, where the thread loop opens for the hook point.
SPI	Stitches per inch; an alternative expression of stitch density (stitches/cm × 2.54).

Key points and summary

- Stitches are classified by ISO 4915 (100–600 classes); seams by ISO 4916 (Classes 1–8). Learn the numbers — buyers' specs are written in them.
- The 301 lockstitch interlaces needle and bobbin threads in mid-ply; loop formation depends on a 1.8–2.2 mm needle rise after bottom dead centre.
- A seam fails by thread rupture, fabric tearing or yarn slippage (grinning); seam efficiency of 70–85% is typical of good work.
- Stitch selection is a stretch decision before it is a strength decision: 301 for stability, 401/504/600 classes where the seam must extend.

Practice

Answer in your own words; brief model answers follow.

1. A seam breaks at 238 N where the fabric breaks at 340 N. Calculate the seam efficiency and comment on it.
2. State the stitch class and full type number for (a) a three-thread overlock, (b) a two-thread chainstitch inseam, (c) a five-thread safety stitch, naming its two component types.
3. In lockstitch formation, what happens if the needle rise after bottom dead centre is set too small?
4. A woven trouser seat seam opens with holes beside the stitching although no thread has broken. Name the failure mode and two production remedies.
5. Why is stitch type 301 unsuitable for the side seam of a stretch-legging, and which classes would you specify instead?

Answer key

1. 238 ÷ 340 × 100 = 70%. Acceptable but at the lower edge of the good range: check thread strength, stitch density and needle damage before accepting the construction.
2. (a) Class 500, type 504. (b) Class 400, type 401. (c) Type 516, combining 401 and 504 sewn simultaneously.
3. The loop does not open fully, so the hook point misses or grazes it: skipped stitches and thread shredding follow. Loop lift must be set to the machine maker's value, about 1.8–2.2 mm.
4. Yarn slippage (seam grinning). Remedies: increase stitch density, use a seam construction with more allowance/lapped seam, specify a stronger thread and check the fabric's slip resistance; widening the seam allowance also helps.
5. Type 301 extends only about 3–5%, so body movement overloads the thread and the seam cracks. Specify chain/overedge/cover classes — 401, 504 or 607 — whose looped structure extends with the

fabric.